Nilay Kushawaha

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Profile Summary

Robotics and AI researcher with a PhD in Biorobotics and postdoctoral experience at New York University Abu Dhabi. My work focuses on continual learning, reinforcement learning, and adaptive control for robotic systems operating in complex, changing environments. I combine theoretical and algorithmic research with hands-on experience deploying intelligent systems in rigid and soft robotics, multimodal perception, computer vision, and sensor fusion. I have published in leading IEEE and Wiley journals and am committed to developing reliable, scalable solutions that move advanced robotics from research into real-world use.

Work Experience

Postdoctoral Associate, New York University Abu Dhabi, UAE *June 2026 – Present*
(Supervisor: Prof. Fares Abu-Dakka)

- Conduct research on continual and reinforcement learning for adaptive robotic control, robot learning, and embodied AI.

Visiting Researcher, National University of Singapore, Singapore *June 2025 – Sept 2025*
(Prof. Cecilia Laschi)

- Developed a dynamic continual learning algorithm to enable localized control of a modular soft robot (MSR) .
- Tested the controller in real-time for trajectory tracking and dynamic reaching tasks, achieving an error less than 6 mm in position and 3° in orientation.

Master's Thesis Project, Jefferson Lab, U.S.A. *July 2021 – July 2022*
(Dr. Yulia Furletova)

- Simulated and modeled the Gas Electron Multiplier (GEM) detector for Electron-Ion Collider project.
- Applied AI algorithm for extracting sensitive particle signals from high-noise environment, improving detection accuracy.

Education

The Biorobotics Institute, Scuola Superiore Sant'Anna, Pisa, Italy *Oct 2022 – May 2026*
PhD in Biorobotics & AI (Supervisor: Prof. Egidio Falotico)

- Credits: 20/20
- Coursework: Machine Learning, Brain Inspired Control, Finite Elements Methods, Deep Learning for Medical Imaging, Robot Programming, Ethics in AI.

Indian Institute of Technology, Indore, India *July 2020 – July 2022*
Master of Science (M.Sc.) in Physics (Supervisor: Prof. Ankhi Roy)

- CGPA: 8.65/10
- Coursework: Mathematical Physics, Statistical Mechanics, Classical Mechanics, Quantum Mechanics, Detector Physics.

University of Delhi, Delhi, India *August 2017 – July 2020*
Bachelor of Science (B.Sc.) in Physics

- CGPA: 8.46/10
- Coursework: Numerical Methods, Introduction to Programming, Computational Physics, Digital/Analog Electronics, Modern Physics.

Skillset

Programming Languages: Python, C++, Scilab, SQL

AI Skills: Machine Learning, Deep Learning, Continual Learning, Reinforcement Learning, Generative AI, Data-driven Control

Platforms & Tools: VSCode, PyTorch, Arduino, Basics of Ethical Hacking, ROS 2, LabVIEW, Franka ROS, Scikit-learn, NumPy, Pandas, GitHub, HTML5, CSS, LaTeX, Lambda 7, Franka Research 3

Soft Skills: Leadership, Teamwork, Adaptability

Training & Certifications

- Advanced Course on Data Science & Machine Learning (Italy, 2024).
- Fundamentals of Deep Learning (NVIDIA, India, 2021).
- First Indian Workshop on Artificial Intelligence (IIT Indore, India, 2021).

Selected Publications and Pre-Prints

- SynapNet: A Complementary Learning System Inspired Algorithm With Real Time Application in Multimodal Perception; **Nilay Kushawaha**, L. Fruzetti, E. Donato, E. Falotico; ***IEEE Transactions on Neural Networks and Learning Systems (2024)***, *Paper link*
- Continual Learning for Multimodal Data Fusion of a Soft Gripper; **Nilay Kushawaha**, E. Falotico; ***Wiley Advanced Robotics Research (2025),*** *Paper link*
- AGPNN: A Dynamic Architecture-Based Continual Reinforcement Learning Algorithm for Robotic Control; **Nilay Kushawaha**, G. Perovic, E. Donato, E. Falotico; ***Accepted(IEEE Transactions on Systems, Man, and Cybernetics: Systems)***
- Adaptive Drift Compensation for Soft Sensorized Finger Using Continual Learning; **Nilay Kushawaha**, R. Pathan, N. Pagliarani, M. Cianchetti, E. Falotico; ***IEEE Robosoft Conference (2025)***, *Paper link*
- Domain Translation of a Soft Robotic Arm Using Conditional Cycle Generative Adversarial Network; **Nilay Kushawaha**, C. Alessi, L. Fruzetti, E. Falotico,; ***IEEE International Conference on Robotic Systems and Applications (2025)***, *Paper link*
- A Continual Learning Framework for Adaptive Control of Modular Soft Robots; **Nilay Kushawaha**, M. S. Nazeer, B. S. Bal, C. Laschi, E. Falotico; ***Under Review (IEEE Access)***, *Preprint*
- LLM-Guided Compositional Continual Learning for Robot Manipulation with Gated Low-Rank Experts; **Nilay Kushawaha**, A. Swikir, F. Abu Dakka; ***Submitted to ICRA 2027***
- SIGMA: Statistical Generative Memory for Dual-Incremental Myoelectric Hand-Pose Decoding; **Nilay Kushawaha**, I. Hmida, A. Swikir, F. Abu Dakka; ***Submitted to ICRA 2027***
- SOFT-VLA: Vision-Language-Action Policy for Proprioception-free Soft Robot Control and Continual Manipulation; **Nilay Kushawaha**, C. Alessi, E. Falotico, L. Natale, F. Abu Dakka; ***Submitted to IEEE Robosoft 2027***

Organizational & Supervision Activities

- Supervised 6 students on a project for the course "Introduction to Robotics & AI", taught by Dr. Falotico.
- Created a four-hour tutorial video on "Advancements in Continual Learning for Robotics" for the Ebrains-Italy project.
- Student Coordinator, Dept. of Physics, IIT Indore.
- Robotics Club Coordinator, University of Delhi.

Achievements & Awards

- Recipient of Enfield AI Scholarship 2025 for exchange.
- Recipient of full PhD scholarship at Scuola Superiore Sant'Anna.
- Recipient of Ishan Uday undergraduate scholarship for 3 consecutive years.
- Undergraduate 2nd semester examination topper.

Academic Service

- Associate Editor, ICRA 2027
- Reviewer for ICRA 2026, IROS 2025
- Reviewer for IEEE TNNLS, TSMCS
- Reviewer for Springer Scientific Reports, Virtual Reality, Soft Computing

Side Project: Calorie Tracking App

CalCal

- Developed a mobile application that estimates calorie content from food images, barcode scans, or manual inputs.
- Implemented features to log and monitor calories burned through selected physical activities.
- Designed an intuitive dashboard summarizing daily and weekly calorie intake and expenditure.